Korea Technology and Information Promotion Agency for SMEs

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Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

Proposer	Name	Deok-Ho Kim	Email	tipa@jhu.edu
Institute Name	Johns Hopkins University School of Medicine			
Address	Rm. 724, Ross Research Building 720 Rutland Avenue Baltimore, MD 21205			

2. Technology Overview

Title of Proposed Technology	AI-Based Remote and Home Mental Health Care Platform for Continuous Monitoring and Personalized Recommendation Services			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display ( ), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI (√), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( )			
Industrial Technology Classification [1]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Bio・Medical		Knowledge Services	
	Medium Category	Small Category	Medium Category	Small Category
	Medical Information & Systems	Remote and Home Health care Devices	R&D / Engineering Services	Testing/ Inspection / Analysis Techniques

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Concept of Proposed Technology	The proposed technology is an **AI-based remote and home mental health care platform** designed to support continuous monitoring and personalized care-support services in non-clinical environments. The platform is intended for use in **remote and home-based settings** and focuses on structured data collection, analysis, and reporting to assist mental health care support activities.
	
	The platform **does not provide diagnosis, therapy, or clinical treatment**. Instead, it supports mental health care by enabling longitudinal monitoring of behavioral and emotional patterns and by delivering data-driven, personalized recommendation content.
	
	Data Collection Framework
	
	The platform continuously collects **real-world mental health–relevant data** through smartphones and wearable devices using **Ecological Momentary Assessment (EMA)** methodologies:
	
	• **Active EMA:**
	  ○ Self-reported mood states
	  ○ Brief standardized assessment items
	• **Passive EMA:**
	  ○ Sleep patterns
	  ○ Physical activity
	  ○ Heart rate–related metrics
	  ○ Smartphone usage logs
	
	AI and LLM-Based Analysis
	
	Multi-modal EMA data are analyzed using **AI models optimized for longitudinal pattern learning and individual baseline–based change detection**. These models identify deviations from personal baselines over time rather than applying population-level diagnostic thresholds.
	
	In addition, **Large Language Models (LLMs)** are employed to transform AI-derived outputs into **human-readable summaries and structured care-support reports**. These LLM-generated outputs are designed to improve interpretability for users and healthcare professionals while **explicitly avoiding autonomous clinical judgment or decision-making**.
	
	International Deployment and Closed-Loop Operation
	
	Within the scope of the international joint R&D project, the platform will be deployed in **overseas remote and home environments** through collaboration with partner hospitals. Data generated through overseas research studies will be applied to the existing platform to update and optimize AI models, personalized recommendation logic, and reporting formats.
	
	The platform operates as a **closed-loop system**, integrating EMA-based data collection, AI/LLM-driven analysis, personalized recommendation delivery, and structured reporting for individuals and healthcare professionals. This closed-loop operation supports continuous improvement of platform robustness and applicability across diverse populations and living environments.

Rationale for Joint R&D Project	**International Deployment and Closed-Loop Operation** Within the scope of the international joint R&D project, the platform will be deployed in **overseas remote and home environments** through collaboration with partner hospitals. Data generated through overseas research studies will be applied to the existing platform to update and optimize AI models, personalized recommendation logic, and reporting formats. The platform operates as a **closed-loop system**, integrating EMA-based data collection, AI/LLM-driven analysis, personalized recommendation delivery, and structured reporting for individuals and healthcare professionals. This closed-loop operation supports continuous improvement of platform

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	robustness and applicability across diverse populations and living environments.
Key Words	1. Remote and Home Mental Health Care 2. Artificial Intelligence 3. Ecological Momentary Assessment 4. Personalized Care Support 5. International Deployment

3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	**Start of the Project**	Level (4)	**End of the Project**	Level (6)
R&D Objectives	• Prepare an existing AI-based remote and home mental health care platform for international deployment through regulatory and operational preparation • Design and conduct overseas research studies to generate real-world EMA datasets in remote and home setting • Apply overseas datasets to the existing platform and evaluate data compatibility and AI model performance • Update and optimize AI models, personalized recommendation logic, and reporting structures using overseas data • Demonstrate feasibility and real-world applicability of the optimized platform to support international market expansion			
R&D Target Specifications				

Category	Target Specification
Data Collection	Continuous real-world EMA data collected via smartphones and wearable devices in remote and home settings
Analysis Model	AI-based longitudinal pattern analysis and individual baseline–based change detection models
Personalization	Data-driven personalized care-support recommendation and monitoring logic
Delivery Channel	Smartphone-based recommendation content and structured reports for individuals and healthcare professionals
Clinical Grounding	Monitoring indicators and reporting structures refined based on hospital-led research findings
Evaluation	Feasibility, usability, and real-world applicability assessment in overseas remote and home environments

R&D Plan	*Phase 1: Regulatory Preparation and Research Design* This phase focuses on preparatory activities prior to overseas study execution. The Korean SME conducts regulatory review and preparation for international deployment, including assessment of applicable digital health regulations, data protection requirements, and operational considerations for remote and home use. The existing platform architecture and data pipelines are reviewed to identify necessary adaptations. In parallel, the overseas research institute and collaborating hospital prepare feasibility and validation

	studies, including protocol design, definition of inclusion and exclusion criteria, selection of monitoring indicators and outcome measures, and acquisition of ethical approvals under local regulations. No data collection occurs during this phase. *Key Deliverables:* • Regulatory readiness and gap analysis report • Overseas research study protocol and ethical approval documentation • Data architecture and AI/LLM update strategy document *Phase 2: Overseas Research Execution and Data Application* This phase constitutes the core joint R&D activities. The overseas hospital conducts the research study and collects real-world Active and Passive EMA data in remote and home environments. Data quality and ethical compliance are ensured throughout the study. Concurrently, the Korean SME applies overseas datasets to the existing platform, evaluates AI model performance across different population characteristics, and performs initial updates to AI models and personalized recommendation logic. Updated AI/LLM outputs are jointly reviewed by technical and clinical experts to ensure interpretability and appropriateness. *Key Deliverables:* • Overseas real-world EMA dataset • AI model and recommendation logic update results • Joint technical–clinical review report - *Phase 3: Platform Optimization and Market Expansion Preparation* This phase focuses on optimization and scaling based on findings from Phase 2. The Korean SME refines AI models, personalized recommendation content, and reporting formats to improve robustness and usability. The overseas hospital provides interpretation of research findings and feedback from clinical and user perspectives. This phase supports transition from a research-validated platform to a **market-ready solution** demonstrated in relevant remote and home healthcare environments. *Key Deliverables:* • Optimized AI-based remote and home mental health care platform • Feasibility and real-world applicability evaluation report • International deployment readiness documentation

4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	• Lead design and execution of overseas feasibility and validation studies • Conduct participant recruitment and real-world data collection • Ensure ethical compliance and data quality • Provide research findings and clinical interpretation to inform platform optimization

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Expected R&R of Korean SME Partner	• Provide and maintain the existing AI-based remote and home mental health care platform • Conduct regulatory preparation and overseas deployment planning • Integrate updated AI/LLM components and refined recommendation content • Prepare for international commercialization and market expansion

5. Expected Outcomes/Impact

Expected Outcomes and Impact	The expected outcome of this project is an **internationally validated and data-optimized AI-based remote and home mental health care platform**, demonstrated in real-world overseas environments. By leveraging overseas research data, the platform will achieve improved robustness, personalization accuracy, and applicability across diverse populations. These outcomes support regulatory readiness and enable scalable international expansion of remote and home mental health care devices. From an industrial perspective, the project strengthens global competitiveness of Korean digital mental health technologies and contributes to sustainable AI-driven solutions that extend mental health care support beyond traditional clinical settings.

6. References

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(2) Murray, Elizabeth, Eric B. Hekler, Gerhard Andersson, Linda M. Collins, Aiden Doherty, Chris Hollis, Daniel E. Rivera, Robert West, and Jeremy C. Wyatt. "Evaluating digital health interventions: key questions and approaches." *American journal of preventive medicine* 51, no. 5 (2016): 843-851.

(3) Peyroteo, Mariana, Inês Augusto Ferreira, Luís Brito Elvas, João Carlos Ferreira, and Luís Velez Lapão. "Remote monitoring systems for patients with chronic diseases in primary health care: systematic review." *JMIR mHealth and uHealth* 9, no. 12 (2021): e28285.

(4) Topol, Eric J. "High-performance medicine: the convergence of human and artificial intelligence." *Nature medicine* 25, no. 1 (2019): 44-56.

(5) WHO. *WHO guideline: recommendations on digital interventions for health system strengthening*, 2019.